Consistency of the density treatment in the compressible Spalart–Allmaras model of OpenFOAM v2412

Paper-to-code comparison and identification of the missing density-gradient term

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August 11, 2026

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1 Summary and principal finding

This report audits the single-phase compressible Spalart–Allmaras (SA) transport equation assembled by OpenFOAM v2412 (which is unchanged to the, as of writing, latest v2606). It compares the native finite-volume equation with the material-form equation cited by OpenFOAM from Spalart and Allmaras (1994) and with its equivalent density-conservative form given by Allmaras, Johnson, and Spalart (2012) [1, 3].

The issue is confined to the diffusion term. Define

$$D_{\widetilde{\nu}} = \frac{\nu + \widetilde{\nu}}{\sigma}.$$

Native OpenFOAM uses the density-weighted diffusive flux

$$\boldsymbol{\nabla} \cdot \left(\rho D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right),$$

whereas multiplying the cited material equation by density requires

$$\rho \boldsymbol{\nabla} \cdot \left(D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu} \right).$$

These expressions are not equal for variable density. The product rule gives

$$\boldsymbol{\nabla} \cdot (\rho D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu}) = \rho \boldsymbol{\nabla} \cdot (D_{\widetilde{\nu}} \boldsymbol{\nabla} \widetilde{\nu}) + D_{\widetilde{\nu}} \boldsymbol{\nabla} \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}.$$

The 2012 conservative equation cancels the final cross-gradient contribution explicitly; native OpenFOAM does not. Consequently, expanding the native equation back into material form produces the additional term

$$\boxed{\frac{\nu + \widetilde{\nu}}{\sigma} \boldsymbol{\nabla} \ln \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}} \tag{1}$$

relative to the cited 1994 equation. Equivalently, relative to the 2012 conservative equation, the native physical right-hand side lacks

$$\boxed{-\frac{\nu + \widetilde{\nu}}{\sigma} \boldsymbol{\nabla} \rho \cdot \boldsymbol{\nabla} \widetilde{\nu}}. \tag{2}$$

- **Single identified mismatch.** Native OpenFOAM places ρ inside the SA diffusive flux without the cross-gradient cancellation required to recover the cited material equation.
- **Two equivalent checks.** The native material equation has the additional positive term in Eq. (1); equivalently, its conservative physical right-hand side lacks the negative term in Eq. (2). These are two descriptions of the same continuum mismatch, not two independent inconsistencies.
- **Corrected implementation.** The custom `SpalartAllmarasDensityCorrected` model adds the required term with the coefficient and matrix sign needed to cancel the native density-product term. It matches the 2012 diffusion coefficient when `sigmaNut` is identified with the paper's σ.
- **Scope.** The mismatch vanishes for constant density or when $\boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\widetilde{\nu} = 0$. The derivation concerns the continuum transport/diffusion sector; it does not establish complete closure identity, exact discrete equivalence, or improved predictions.
- **Independent NASA implementation warning.** Note 4 of the NASA Turbulence Modeling Resource warns that some SA implementations "incorrectly introduce the density inside the derivatives of the SA equation" and states that this changes predictions for supersonic flows; it explicitly distinguishes the separately formulated Catris–Aupoix proposal as having a valid basis [4]. The NASA note does not identify OpenFOAM by name. Its relevance here follows from the independent reconstruction of Eq. (14), which establishes that the native implementation has precisely the warned-about derivative structure.

Outline

- Section 2 identifies the literature equations, citation basis, and audit scope.
- Section 3 follows the actual OpenFOAM operators from code to the native OpenFOAMimplemented conservative continuum equation.
- Section 4 directly compares the Allmaras 2012 and native OpenFOAM conservative equations and identifies the missing term.
- Section 5 summarizes the established algebraic result and interprets the native equation as a modified SA model.
- Appendix A audits the custom correction, including the `fvm::Su` sign and a dimensional check.
- Appendix B retains the complete line-by-line derivation from the material equation to its density-conservative form.

2 Reference equations and audit scope

The SA working variable is denoted by $\widetilde{\nu}$, the molecular kinematic viscosity by ν, density by ρ, velocity by $\boldsymbol{u}$, and the model diffusion constant by σ. The production, destruction, and trip contributions are grouped as P, D, and T, respectively. The diffusion coefficient is left explicitly as $(\nu + \widetilde{\nu})/\sigma$ so that every factor remains visible.

2.1 The cited 1994 material formulation

The audited OpenFOAM v2412 header cites the 1994 journal article by Spalart and Allmaras as the standard-model reference. That article explicitly defines $\mathrm{D}/\mathrm{D}t$ as the Lagrangian or material derivative and gives the full Version-Ia working-variable equation in Appendix Eq. (A2), printed on p. 20 (PDF page 16):

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu} \\ + \frac{1}{\sigma}\left\{\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + c_{b2}\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\right\} \\ - \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left(\frac{\widetilde{\nu}}{d}\right)^2 + f_{t1}(\Delta U)^2. \end{aligned} \tag{3}$$

Production, destruction, and tripping differ in detail from the implementation audited below, but the material transport and diffusion structure is the one relevant to the density comparison.

The final paragraph of the 1994 appendix states that Eqs. (A1)–(A6) had been applied directly in compressible flow and reports only a weak influence of density variations in the slightly supersonic boundary layers then examined. Thus the cited journal article supports a density-free material equation for compressible use. It does not give a density-conservative finite-volume recasting or prescribe where ρ should be placed within such a recasting. The Turbulence Modeling Resource identifies this journal article as the official publication arising from the 1992 conference paper and states that the journal version takes precedence [1, 4].

2.2 The Allmaras 2012 standard compressible form

This resource explicitly states a conservative form of the SA model, which can be directly compared to the OpenFOAM implementation. For this Allmaras et al. (2012) first write their transport equation Eq. (2), printed on p. 2, as

$$\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = P - D + T + \frac{1}{\sigma}\left\{\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + c_{b2}\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\right\}. \tag{4}$$

where production, wall destruction and trip terms are

$$P = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu}, \qquad D = \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left[\frac{\widetilde{\nu}}{d}\right]^2, \qquad T = f_{t1}(\Delta u)^2.$$

This formulation is equivalent to the original 1994 model equation (see 3)

Their Section 2.2 explicitly states that this equation applies to both incompressible and compressible flow and should be treated as the standard compressible form. They then give the equivalent conservation form as their Eq. (9), printed on p. 3:

density conservative SA model equation

$$\begin{aligned} 0 = \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P-D+T) \\ - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2 \\ + \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{5}$$

The final term is positive because Eq. (5) is an all-terms-left residual. It is negative when moved to the physical right-hand side. Allmaras 2012 Eq. (10), their shock-layer integral, independently repeats the negative right-hand-side sign [3].

Appendix B contains the full mathematical derivation from the material equation Eq. (3) to the density conservative form Eq. (5) and proofs that no density external correction or similar is being applied, but that it is a clean mathematical derivation. Thus Eq. (5) should be understood as the faithful density conservative model equation of the original SA model.

2.3 The 1992 Outlook on possible compressibility additions

The earlier conference paper is retained here only for its historical Outlook. That discussion mentions possible compressibility additions involving $\boldsymbol{\nabla}\rho \cdot \boldsymbol{\nabla}\widetilde{\nu}$, considers writing the transport equation for $\widetilde{\mu}$ instead of $\widetilde{\nu}$, and notes freedom to place density inside or outside derivatives. It does not prescribe a final coefficient, sign, or unique conservative equation, and therefore does not define the density-weighted flux used by native OpenFOAM or establish its equivalence to the standard material equation [2].

3 Reconstruction of the native OpenFOAM equation

The governing equation is assembled in the shared SA base class. The relevant OpenFOAM v2412 code is shown first and subsequently the continuum equation is translated from it term-by-term to the underlying mathematical equation. Which can then be compared to the Allmaras 2012 published density conservative SA model equation.

Listing 1: Native SA equation assembly in SpalartAllmarasBase.C.

```
468 fvm::ddt(alpha, rho, nuTilda_)
469 + fvm::div(alphaRhoPhi, nuTilda_)
470 - fvm::laplacian(alpha*rho*DnuTildaEff(), nuTilda_)
471 - Cb2_/sigmaNut_*alpha()*rho()*magSqr(fvc::grad(nuTilda_)())
472 ==
473   Cb1_*alpha()*rho()*Stilda()*nuTilda_()*(1 - ft2())
474 - fvm::Sp
475   (
476     (Cw1_*fw(Stilda, dTilda) - Cb1_/sqr(kappa_)*ft2())
477    *alpha()*rho()*nuTilda_()/sqr(dTilda()),
478     nuTilda_
479   )
480 + fvOptions(alpha, rho, nuTilda_)
```

This code is at `SpalartAllmarasBase.C:466-481`. The coefficient used by the Laplacian is defined at lines 383–389 as

$$\texttt{DnuTildaEff()} = \frac{\nu + \widetilde{\nu}}{\sigma}. \tag{6}$$

Here and below, the code coefficient `sigmaNut` is denoted by σ. Its v2412 default is the configurable decimal `0.66666`; exact numerical agreement with the paper value $\sigma = 2/3$ requires configuring the same value. The custom correction reuses `DnuTildaEff` and therefore cancels the native density-product term for either setting.

The thin RAS wrapper does not replace this equation; it supplies wall distance and viscosity-related updates and inherits the base-class transport equation.

The reconstruction below is restricted to the bulk/interior transport equation assembled in Listing 1. It precedes and does not include the subsequent `relax()`, `fvOptions.constrain()`, `boundaryManipulate()` (including any boundary-row edits made by constrained patch fields),

`solve()`, the post-solve `bound()`, or `nuTilda_.correctBoundaryConditions()` calls. Those operations may modify matrix coefficients, sources, boundary-adjacent rows, or the accepted field, but in an audit it was not found that these operations correct the identified model deviation.

3.1 Translate each code term into continuum notation

For the registered single-phase compressible model, $\alpha = 1$, ρ is the density field, and `alphaRhoPhi` is the face mass flux corresponding to $\rho \boldsymbol{u} \cdot \boldsymbol{S}_f$. The mappings below are for this single-phase registration and interpret the displayed matrix as an all-terms-left residual before reporting a physical right-hand-side sign.

3.1.1 Transient and convection operators

With `fvm::ddt` and `fvm::div` read as their canonical unbounded continuum operators, the first two code terms,

$$\begin{aligned}&\texttt{fvm::ddt(alpha, rho, nuTilda_)}\\ &\quad + \texttt{fvm::div(alphaRhoPhi, nuTilda_)},\end{aligned}$$

represent

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}), \tag{7}$$

This is initially a conservative transport operator, not directly the material derivative. For differentiable fields, applying the product rule gives

$$\begin{aligned}&\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu})\\ &\quad = \rho\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} + \widetilde{\nu}\left[\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u})\right].\end{aligned} \tag{8}$$

Mass continuity, $\partial_t\rho + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0$, removes the bracket. Consequently,

$$\texttt{ddt + div} \quad\longrightarrow\quad \rho\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu},$$

and it becomes the unweighted material derivative only after the complete equation is divided by $\rho > 0$.

This is a continuum/operator mapping, not a scheme-independent statement about the assembled matrix. The selected `fvSchemes` may wrap the transient or convection operator in a bounded form. In particular, the bounded three-field time scheme subtracts

$$\texttt{fvm::Sp(fvc::ddt(alpha,rho),vf)},$$

whereas the bounded convection scheme subtracts

$$\texttt{fvm::Sp(fvc::surfaceIntegrate(faceFlux),vf)}.$$

These terms change how a discrete mass-continuity residual is apportioned between transport and source; they do not contain the diffusion cross-gradient $(\nu + \widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}/\sigma$.

3.1.2 Laplacian operator

Using Eq. (6), the matrix-left term

$$-\texttt{fvm::laplacian}\left(\rho\frac{\nu+\widetilde{\nu}}{\sigma},\widetilde{\nu}\right)$$

represents

$$-\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\,. \tag{9}$$

Density is inside the coefficient passed to the Laplacian. The code therefore discretizes a density-weighted conservative diffusive flux; it does not discretize ρ multiplied by a density-free Laplacian.

3.1.3 Explicit c_{b2} term

The code term

$$-\texttt{Cb2_/sigmaNut_*rho()*magSqr(fvc::grad(nuTilda_)()())}$$

maps to

$$-\frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\,. \tag{10}$$

The c_{b2} term is part of the modeled nonlinear diffusion in both papers. It is distinct from the density product-rule term and must not be interpreted as the missing cancellation.

3.1.4 Production and destruction terms

The first term on the code right-hand side represents the paper's production term:

$$\texttt{Cb1_*rho()*Stilda()*nuTilda_()*(1-ft2())} \quad\longrightarrow\quad \rho P,$$

where

$$P = c_{b1}(1-f_{t2})\widetilde{S}\widetilde{\nu}. \tag{11}$$

The implicit `fvm::Sp` term represents destruction. Because it is preceded by a minus sign on the code right-hand side, its continuum meaning is $-\rho D$, where

$$D = \left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left(\frac{\widetilde{\nu}}{d}\right)^2. \tag{12}$$

Thus the modeled right-hand side in the native equation is $\rho(P-D)$. Native OpenFOAM has no trip term T; comparison with the full paper equation therefore uses $T=0$. The final `fvOptions(alpha, rho, nuTilda_)` entry permits additional case-defined sources. It is not part of the SA paper equation and is omitted from the reconstructed native equation.

For the density audit, P and D are held symbolically fixed so that the transport and diffusion sectors can be compared directly. This does not assert complete closure identity: native OpenFOAM uses the $\widetilde{S}$ limiter $\max[\Omega + f_{v2}\widetilde{\nu}/(\kappa d)^2, C_s\Omega]$, which is not identical to the Allmaras 2012 baseline and proposed negative-$\widetilde{S}$ continuation.

3.2 Assemble the mathematical equation

Combining the preceding mappings gives the all-terms-left continuum meaning of the native finite-volume assembly:

$$0 = \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P - D) - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu + \widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\,. \tag{13}$$

Equivalently, moving the modeled and diffusion terms to the physical right-hand side gives:

Native OpenFOAM continuum equation

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) = \rho(P - D) + \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu + \widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + \frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2\,. \tag{14}$$

Equation (14) is not copied from either paper; it is the continuum equation represented by the audited native finite-volume assembly, expressed in the papers' notation with $\alpha = 1$, `fvOptions` suppressed, and $T = 0$. For the incompressible registration, both α and ρ are compile-time one fields, so the density issue vanishes exactly.

4 Direct comparison and identified error

Equation (5) is the Allmaras 2012 conservative equation, and Eq. (14) is the equation reconstructed term by term from native OpenFOAM. Set $T = 0$, as required by the native implementation, and hold P, D, and σ fixed. After moving the modeled and diffusion terms of Eq. (5) to the physical right-hand side, the complete comparison is

$$\mathrm{RHS}|_{\text{Allmaras Eq. (5)},\,T=0} = \mathrm{RHS}|_{\text{OpenFOAM Eq. (14)}} - \frac{\nu + \widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \tag{15}$$

Every other displayed transport, production, destruction, and diffusion term is identical under this comparison. Equation (14) therefore lacks the final negative density-gradient term.

That term is the cancellation required by the product rule

$$\boldsymbol{\nabla}\cdot[\rho(\nu + \widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] = \rho\boldsymbol{\nabla}\cdot[(\nu + \widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + (\nu + \widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

Without the cancellation, rewriting the native equation in material form produces the corresponding positive cross-gradient term.

Identified error in the OpenFOAMimplemented equation.
Relative to Eq. (5), the physical right-hand side of Eq. (14) lacks

$$-\frac{\nu + \widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

Equivalently, the native material equation contains the additional term

$$+\frac{\nu + \widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\ln\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.$$

The equations agree when density is constant, $\boldsymbol{\nabla}\widetilde{\nu} = 0$, or the density and working-variable gradients are orthogonal. A nonzero density gradient by itself is not sufficient; the cross-gradient alignment determines the contribution.

5 Conclusion

Native single-phase compressible OpenFOAM v2412 uses the density-weighted diffusive flux

$$\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]$$

without the explicit conservative cancellation needed to recover

$$\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\,.$$

Its material form therefore contains the additional positive term in Eq. (1). The custom `SpalartAllmarasDensityCorrected` source adds the corresponding positive term to the matrix residual, which is negative on the physical right-hand side and cancels the native product-rule contribution. With the same σ, the resulting diffusion structure agrees with Allmaras 2012 Eq. (9). This cancellation is exact at the continuum level under the assumptions stated during the reconstruction in Section 3; it is not a claim that all nondiffusive closure choices are identical.

Equation (14) nevertheless defines a mathematically consistent conservative scalar-transport model if the diffusive flux

$$\boldsymbol{J}_{\widetilde{\nu}} = -\rho D_{\widetilde{\nu}}\boldsymbol{\nabla}\widetilde{\nu}, \qquad D_{\widetilde{\nu}} \equiv \frac{\nu+\widetilde{\nu}}{\sigma}, \tag{16}$$

is chosen intentionally. That choice, however, generates the additional density-gradient term in the corresponding material equation. Without the cancellation prescribed by Allmaras et al. (2012), the native equation is therefore a modified SA model rather than the intended compressible SA formulation.

The defensible conclusion is thus a *confirmed transport/diffusion-sector deviation from both the 1994 material formulation cited by OpenFOAM and the Allmaras 2012 conservative standard*. This audit cannot determine whether the deviation was intentional or arose from an unintended derivation error. The 1992 Outlook shows that alternative compressibility additions and density placements were contemplated, but it does not define the native density-weighted flux or establish that flux as the intended SA model [1, 2, 3]. The native OpenFOAM equation and the cited standard therefore represent different models in compressible flow.

A Implementation of the corrected model

The custom model first assembles the native equation and then adds the source shown below. The relevant code is in `SpalartAllmarasDensityCorrected.C:54-66,136-175`.

Listing 2: Density-gradient source in SpalartAllmarasDensityCorrected.

```
54 const tmp<volVectorField> tgradRho = fvc::grad(rho);
55
56 // The stock conservative diffusion contributes
57 //     div(rho*DnuTildaEff*grad(nuTilda))
58 // on the physical RHS.  Adding this explicit source to the matrix LHS
59 // subtracts DnuTildaEff*grad(rho).grad(nuTilda) from that RHS, giving
60 // exactly rho*div(DnuTildaEff*grad(nuTilda)).
61 nuTildaEqn += fvm::Su
62 (
63     DnuTildaEff()
64    *(tgradRho()() & gradNuTilda()),
65     this->nuTilda_
66 );
```

The `fvm::Su` convention resolves the apparent sign ambiguity. The explicit calls to the field-internal accessors make the expression in Listing 2 a temporary `DimensionedField`. The overload in `fvmSup.C:92-103` forwards that temporary to the field implementation at lines 72–89, where the supplied field is stored as `matrix.source() -= V*source`. Adding the source therefore adds

$$+\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu} \tag{17}$$

to the all-terms-left matrix residual, or the negative of this expression to the physical right-hand side. It has the form and sign of the final term in Eq. (5), is exactly the product-rule cancellation identified in Section 4 for the native `DnuTildaEff`, and has the Allmaras 2012 numerical coefficient when `sigmaNut` is set to 2/3.

The dimensions are consistent:

$$\left[\frac{\nu+\widetilde{\nu}}{\sigma}\right]=L^2T^{-1},\qquad[\boldsymbol{\nabla}\rho]=ML^{-4},\qquad[\boldsymbol{\nabla}\widetilde{\nu}]=LT^{-1}, \tag{18}$$

and hence

$$\left[\frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}\right]=ML^{-1}T^{-2}=\left[\frac{\partial(\rho\widetilde{\nu})}{\partial t}\right]. \tag{19}$$

The custom header provides a no-op overload for `geometricOneField` density, so the incompressible registration remains the native SA equation. The library is registered only in the single-phase compressible and incompressible turbulence tables. Its omission of a phase-fraction-gradient term is consistent with that scope; applying the same source to a variable-α multiphase equation would require a new derivation.

Finally, the modeled term

$$\frac{c_{b2}}{\sigma}\,|\boldsymbol{\nabla}\widetilde{\nu}|^2 \tag{20}$$

is not the density correction. Expanding $\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]$ already retains the derivatives of $\nu+\widetilde{\nu}$, while c_{b2} is a separate model coefficient. The custom cancellation neither removes nor duplicates it.

B Allmaras 2012 material form to conservative form

This section retains the complete conversion from Allmaras 2012 Eq. (2) to Eq. (9). It makes explicit which terms are supplied by mass conservation and which term is created by rewriting the diffusion operator.

The Allmaras 2012 material equation is Eq. (4):

$$\frac{\mathrm{D}}{\mathrm{D}t}\widetilde{\nu} = P - D + T + \frac{1}{\sigma}\left[\boldsymbol{\nabla}\cdot((\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}) + c_{b2}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2\right]. \tag{21}$$

B.1 Put the material equation in zero-residual form

Move every right-hand-side term to the left and expand the material derivative:

$$\begin{aligned}\{\mathrm{SA}\} = \frac{\partial\widetilde{\nu}}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} - (P - D + T)\\ - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2 = 0.\end{aligned} \tag{22}$$

The braces label the complete residual; they introduce no new model quantity.

B.2 Add the zero mass residual

Mass conservation in zero-residual form is

$$\{\mathrm{mass}\} = \frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0. \tag{23}$$

Allmaras et al. (2012) combine the two residuals as

$$0 = \rho\{\mathrm{SA}\} + \widetilde{\nu}\{\mathrm{mass}\}. \tag{24}$$

Because both residuals are zero, multiplying and adding them does not change the solution.

B.3 Multiply out the residuals

Multiplying Eq. (22) by ρ gives

$$\begin{aligned}\rho\{\mathrm{SA}\} = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} - \rho(P-D+T)\\ -\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2.\end{aligned} \tag{25}$$

Multiplying Eq. (23) by $\widetilde{\nu}$ gives

$$\widetilde{\nu}\{\mathrm{mass}\} = \widetilde{\nu}\frac{\partial\rho}{\partial t} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}). \tag{26}$$

Adding Eqs. (25) and (26) yields

$$\begin{aligned}0 = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \widetilde{\nu}\frac{\partial\rho}{\partial t}\\ +\rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) - \rho(P-D+T)\\ -\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2.\end{aligned} \tag{27}$$

B.4 Combine transient and convection products

For the transient terms,

$$\frac{\partial(\rho\widetilde{\nu})}{\partial t} = \rho\frac{\partial\widetilde{\nu}}{\partial t} + \widetilde{\nu}\frac{\partial\rho}{\partial t}. \tag{28}$$

For a scalar $\widetilde{\nu}$ and vector $\rho\boldsymbol{u}$,

$$\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) = \rho\boldsymbol{u}\cdot\boldsymbol{\nabla}\widetilde{\nu} + \widetilde{\nu}\boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}). \tag{29}$$

Substitution into Eq. (27) gives

$$\begin{aligned}0 = \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P-D+T)\\ -\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\left|\boldsymbol{\nabla}\widetilde{\nu}\right|^2.\end{aligned} \tag{30}$$

Only the diffusion term remains to be rewritten.

B.5 Rewrite the diffusion term

Apply the vector product rule:

$$\begin{aligned}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\\ = \rho\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + (\nu+\widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.\end{aligned} \tag{31}$$

Solve for the density-multiplied diffusion that appears in Eq. (30):

$$\begin{aligned}\rho\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\\ = \boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - (\nu+\widetilde{\nu})\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.\end{aligned} \tag{32}$$

The term in Eq. (30) is negative, so

$$\begin{aligned}-\frac{\rho}{\sigma}\boldsymbol{\nabla}\cdot[(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}]\\ = -\frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu+\widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] + \frac{\nu+\widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}.\end{aligned} \tag{33}$$

The positive final sign follows from two negative signs: the one already in front of the residual diffusion and the one introduced by rearranging the product rule.

B.6 Recover Allmaras 2012 Eq. (9)

Substituting Eq. (33) into Eq. (30) gives

$$\begin{aligned} 0 = & \frac{\partial(\rho\widetilde{\nu})}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}\widetilde{\nu}) - \rho(P - D + T) \\ & - \frac{1}{\sigma}\boldsymbol{\nabla}\cdot[\rho(\nu + \widetilde{\nu})\boldsymbol{\nabla}\widetilde{\nu}] - \frac{c_{b2}}{\sigma}\rho\,|\boldsymbol{\nabla}\widetilde{\nu}|^2 \\ & + \frac{\nu + \widetilde{\nu}}{\sigma}\boldsymbol{\nabla}\rho\cdot\boldsymbol{\nabla}\widetilde{\nu}. \end{aligned} \tag{34}$$

This is Allmaras 2012 Eq. (9). The mass residual creates conservative storage and convection. The diffusion product rule, not continuity, creates the final density-gradient term.

Audited source map and implementation notes

The line numbers below refer to the source snapshot used for this appendix and should be rechecked if the upstream or custom source is changed.

Purpose	Source path and audited lines
Native effective diffusivity	`/opt/of-v2412/src/TurbulenceModels/turbulenceModels/Base/SpalartAllmaras/SpalartAllmarasBase.C`, lines 381–390.
Native transport equation	Same file, lines 440–493; equation assembly at 466–481.
Native post-assembly operations	Same file, lines 483–489 for relaxation, option constraints, `boundaryManipulate()`, solve, post-solve bounding, and boundary correction.
Native $\widetilde{S}$ continuation	Same file, lines 141–156.
Native σ default	Same file, lines 206–213.
Native RAS wrapper	`/opt/of-v2412/src/TurbulenceModels/turbulenceModels/RAS/SpalartAllmaras/SpalartAllmaras.C`, lines 40–95.
Single-phase compressible types	`/opt/of-v2412/src/TurbulenceModels/compressible/CompressibleTurbulenceModel/CompressibleTurbulenceModel.H`, lines 68–80 and 120–130.
Custom correction	`SpalartAllmarasDensityCorrected/SpalartAllmarasDensityCorrected.C`, helper at lines 44–67 and input/call path at lines 136–175.
Custom post-assembly operations	Same file, lines 177–193 for relaxation, option constraints, `boundaryManipulate()`, solve, post-solve bounding, and boundary correction.
Custom incompressible no-op	`SpalartAllmarasDensityCorrected/SpalartAllmarasDensityCorrected.H`, lines 77–94.
Custom runtime registrations	`SpalartAllmarasDensityCorrected/makeCompressibleSpalartAllmarasDensityCorrected.C` and `SpalartAllmarasDensityCorrected/makeIncompressibleSpalartAllmarasDensityCorrected.C`, lines 13–16 in each; both translation units are listed in `SpalartAllmarasDensityCorrected/Make/files`.
Explicit-source convention	`/opt/of-v2412/src/finiteVolume/finiteVolume/fvm/fvmSup.C`, lines 92–103 for temporary-field forwarding and lines 72–89 for the matrix-source insertion.
Bounded transient correction	`/opt/of-v2412/src/finiteVolume/finiteVolume/ddtSchemes/boundedDdtScheme/boundedDdtScheme.C`, lines 140–152 for the three-field `fvmDdt(alpha,rho,vf)` overload.
Bounded convection correction	`/opt/of-v2412/src/finiteVolume/finiteVolume/convectionSchemes/boundedConvectionScheme/boundedConvectionScheme.C`, lines 76–87 for `fvmDiv(faceFlux,vf)`.

Two implementation notes delimit the source evidence:

- Immediately below Allmaras 2012 Eq. (9), the paper appears to print $\mu = \nu/\rho$, although its Eq. (1) defines ν as kinematic viscosity. The dimensionally conventional relation is $\mu = \rho\nu$. This apparent typographical error is not used in the derivation and does not change Allmaras 2012 Eq. (9).
- The examined native source contains a local `boundaryManipulate()` hook in the SA solution path. The density-weighted transport equation audited here is otherwise unchanged relative to that source tree's Git `HEAD`.

References

[1] P. R. Spalart and S. R. Allmaras, "A One-Equation Turbulence Model for Aerodynamic Flows," *La Recherche Aérospatiale*, No. 1, pp. 5–21, 1994. Journal version available from the Turbulence Modeling Resource archive.

[2] P. R. Spalart and S. R. Allmaras, "A One-Equation Turbulence Model for Aerodynamic Flows," AIAA Paper 92-0439, 30th Aerospace Sciences Meeting and Exhibit, 1992.

[3] S. R. Allmaras, F. T. Johnson, and P. R. Spalart, "Modifications and Clarifications for the Implementation of the Spalart–Allmaras Turbulence Model," Seventh International Conference on Computational Fluid Dynamics, ICCFD7-1902, 2012.

[4] NASA Langley Research Center, "The Spalart–Allmaras Turbulence Model," Turbulence Modeling Resource, https://tmbwg.github.io/turbmodels/spalart.html, accessed September 1, 2026.